higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE

MATHEMATICS N5

(16030175)

25 November 2025 (X-paper)
09:00–12:00

Scientific calculators and drawing instruments may be used.

This question paper consists of 5 pages and a formula sheet of 5 pages.

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
MATHEMATICS N5
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer all the questions.
2. Read all the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Show all intermediate steps and simplify where possible.
5. All final answers must be rounded off to THREE decimal places.
6. Questions may be answered in any order, but subsections of questions must be kept together.
7. Sketches must be large, neat and fully labelled.
8. Use only a black or blue pen.
9. Write neatly and legibly.

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QUESTION 1

1.1 Determine the following limit:

$$y = \lim_{x \to 0} \frac{3x^2 + e^x}{2x^3 + 5x} \quad (4)$$

1.2 Determine the value(s) of x in the domain $(0; 360°)$ for which $f(x)$ is discontinuous if:

$$f(x) = \sec x \quad (2)$$

[6]

QUESTION 2

2.1 Determine $\frac{dy}{dx}$ in each of the following cases:

(Simplification NOT required)

2.1.1 $y = \ln^3 \left[x^2 - 5x + \sqrt{x^2 - 7x + 20}\right]$ (5)

2.1.2 $y = \sec^{-1}(x^2) + \tan^{-1}\left(\frac{1-x}{1+x}\right)$ (5)

2.2 Calculate $\frac{dy}{dx}$ if $y = 3^{x \ln x^2}$ with aid of logarithmic differentiation. (4)

2.3 Given the implicit function:

$$x^2 y + \cos\left(\frac{\pi}{2} y\right) - 4x^2 = 5$$

2.3.1 Determine $\frac{dy}{dx}$. (3)

2.3.2 Determine the equation of the tangent to the graph at the point $(1, 2)$. (3)

[20]

QUESTION 3

3.1 Given: $f(x) = x^3 - 3x + 5$

3.1.1 Determine the coordinates of the turning points of $f(x)$. (3)

3.1.2 Draw up a table of x and $f(x)$, where x is ranging from $x = -3$ to $x = 2$. (2)

3.1.3 Draw a neat graph of $f(x)$ between these values, showing the turning points on it. (2)

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3.1.4 Given that $x = -2{,}3$ is an approximate root of the equation $x^3 - 3x + 5 = 0$, use Taylor's /Newton's method to determine a better first and second approximation of this root. (Root correct to THREE decimal figures.) (4)

3.2 A particle moves along a horizontal line according to $s = \frac{1}{3}t^3 - 2t^2 + 3t$ where s is the displacement (in metres) at time t (in seconds).

3.2.1 Calculate the displacement of the particle after *2 s*. (2)

3.2.2 Find the velocity at time t and after 4 s. (2)

3.2.3 At what times is the particle at rest? (3)

3.2.4 Find the acceleration at time t and after 4 s. (2)

[20]

QUESTION 4

4.1 Determine $\int y\,dx$ in each of the following cases:

4.1.1 $y = \dfrac{x(2-3x)}{e^{3x} + x^2}$

HINT: let $u = 1 + x^2e^{-3x}$ (3)

4.1.2 $y = \cos^7 x \sin^3 x$ (4)

4.1.3 $y = \cos\left(\frac{\pi}{2}x\right)\sin\left(\frac{\pi}{4}x\right)$ (3)

4.1.4 $y = \dfrac{2x^3 - 5x^2 + 6x + 12}{x - 4}$ (5)

4.1.5 $y = 4x\cos 4x$ (5)

4.2 Determine $\int y\,dx$ by resolving the integral into partial fractions:

$$y = \frac{x-5}{x^2 + 5x + 4}$$ (5)

[24]

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QUESTION 5

5.1 Determine by integration:

$$\int_0^{\infty} e^{-st} \cdot f(t)\,dt \;\; if \; f(t) = \frac{7}{3t^{-1}}$$ (5)

5.2 Given: $x + y = 1$ and $y = \sqrt{1 - x^2}$

5.2.1 Calculate the coordinates of the points of intersection. (3)

5.2.2 Make a neat sketch to show the enclosed area, the representative strip and the point of intersection. (2)

5.2.3 Calculate the magnitude of the area in QUESTION 5.2.2. (3)

5.2.4 Calculate the volume of the solid of revolution formed when the area in QUESTION 5.2.2 is rotated about the x-axis. (4)

5.3 Calculate the moment of inertia of a circular lamina of radius r and mass m about an axis through its centre and perpendicular to the plane of the lamina. (5)

[22]

QUESTION 6

6.1 Determine the particular solution of:

$$\frac{dy}{dx} = \frac{3}{y\cos^2 x}, \qquad \text{if } x = 45° \text{ and } y = 2$$ (5)

6.2 Determine the general solution of:

$$\cos x \frac{d^2y}{dx^2} = \sin 2x + 2^{3x} \cdot \cos x$$ (3)

[8]

TOTAL: 100

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FORMULA SHEET

Any applicable formula may also be used

TRIGONOMETRY

$\sin^2 x + \cos^2 x = 1$

$1 + \tan^2 x = \sec^2 x$

$1 + \cot^2 x = \operatorname{cosec}^2 x$

$\sin 2A = 2 \sin A \cos A$

$\cos 2A = \cos^2 A - \sin^2 A$

$$\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$$

$\sin^2 A = {}^1/_2 - {}^1/_2 \cos 2A$

$\cos^2 A = {}^1/_2 + {}^1/_2 \cos 2A$

$\sin(A \pm B) = \sin A \cos B \pm \sin B \cos A$

$\cos(A \pm B) = \cos A \cos B \pm \sin A \sin B$

$$\tan(A \pm B) = \frac{\tan A \pm \tan B}{1 \pm \tan A \tan B}$$

$\sin A \cos B = {}^1/_2 [\sin(A + B) + \sin(A - B)]$

$\cos A \sin B = {}^1/_2 [\sin(A + B) - \sin(A - B)]$

$\cos A \cos B = {}^1/_2 [\cos(A + B) + \cos(A - B)]$

$\sin A \sin B = {}^1/_2 [\cos(A - B) - \cos(A + B)]$

$$\tan x = \frac{\sin x}{\cos x}; \sin x = \frac{1}{cosec\ x}; \cos x = \frac{1}{\sec x}$$

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BINOMIAL THEOREM

$$(x+h)^n = x^n + \mathrm{n}x^{n-1}\,h + \frac{n(n-1)}{2!}x^{n-2}\,h^2 + \cdots$$

DIFFERENTIATION

$$e = -\frac{f(a)}{f'(a)}$$

$$r = a + e$$

PRODUCT RULE

$$\mathrm{y} = \mathrm{u}(x)\,.\,v(x)$$

$$\frac{dy}{dx} = u\,.\frac{dv}{dx} + v\,.\frac{du}{dx}$$

$$= u\,.v' + v\,.u'$$

QUOTIENT RULE

$$y = \frac{u(x)}{v(x)}$$

$$\frac{dy}{dx} = \frac{v\,.\,\frac{du}{dx} - u\,.\,\frac{dv}{dx}}{v^2}$$

$$\frac{v\,.\,u' - u\,.\,v'}{v^2}$$

CHAIN RULE

$$y = f\big(u(x)\big)$$

$$\frac{dy}{dx} = \frac{dy}{du}\,.\frac{du}{dx}$$

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$f(x)$	$\frac{d}{dx}f(x)$	$\int f(x)dx$
ax^n	nax^{n-1}	$\frac{ax^{n+1}}{n+1}+c$
a	0	$ax+c$
e^x	e^x	e^x+c
a^x	$a^x.lna$	$\frac{a^x}{lna}+c$
$\log_e x$	$\frac{1}{x}$	——
$\log_a x$	$\frac{1}{xlna}$	——
$\sin x$	$\cos x$	$-\cos x+c$
$\cos x$	$-\sin x$	$\sin x+c$
$\tan x$	$\sec^2 x$	$ln(\sec x)+c$
$\cot x$	$-\operatorname{cosec}^2 x$	$ln(\sin x)+c$
$\sec x$	$\sec x\,.\tan x$	$ln[\sec x+\tan x]+c$
$\operatorname{cosec} x$	$-\operatorname{cosec} x\,.\cot x$	$ln(\operatorname{cosec} x-\cot x)+c$
$\sin^{-1} x$	$\frac{1}{\sqrt{1-x^2}}$	——
$\cos^{-1} x$	$\frac{-1}{\sqrt{1-x^2}}$	——
$\tan^{-1} x$	$\frac{1}{1+x^2}$	——
$\cot^{-1} x$	$\frac{-1}{1+x^2}$	——
$\sec^{-1} x$	$\frac{1}{\sqrt{x^2-1}}$	——
$\operatorname{cosec}^{-1} x$	$\frac{-1}{x\sqrt{x^2-1}}$	——

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$f(x)$	$\frac{d}{dx}f(x)$	$\int f(x)dx$
$\frac{1}{\sqrt{a^2-x^2}}$	—	$\sin^{-1}\left(\frac{x}{a}\right)+c$
$\frac{1}{a^2+x^2}$	—	$\frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right)+c$
$\frac{1}{x\sqrt{x^2+a^2}}$	—	$\frac{1}{a}\sec^{-1}\left(\frac{x}{a}\right)+c$
$\sqrt{a^2-x^2}$	—	$\frac{a^2}{2}\sin^{-1}\left(\frac{x}{a}\right)+\frac{x}{2}\sqrt{a^2-x^2}+c$
$\frac{1}{x^2-a^2}$	—	$\frac{1}{2a}ln\left(\frac{x-a}{x+a}\right)+c$
$\frac{1}{a^2-x^2}$	—	$\frac{1}{2a}ln\left(\frac{a+x}{a-x}\right)+c$

INTEGRATION

$$\int f(x)\,g'(x)\,dx = f(x)\,g(x) - \int f'(x)g\,(x)dx$$

$$\int \frac{f'(x)}{f(x)}dx = ln\,f(x)\,+c$$

$$\int [f(x)]^n\,f'(x)dx = \frac{[f(x)]^{n+1}}{n\,+1}+c$$

$$\frac{f(x)}{(ax+b)(cx\,+d)} = \frac{A}{ax\,+b}+\frac{B}{cx\,+d}$$

$$\frac{f(x)}{(x\,+a\,)^n} = \frac{A}{(x\,+a)}+\frac{B}{(x\,+a)^2}+\frac{C}{(x+a)^3}+\cdots\frac{Z}{(x\,+a)^n}$$

APPLICATIONS OF INTEGRATION

AREAS

$$A_x = \int_a^b ydx\;;\; A_x = \int_a^b (y_1 - y_1)dx$$

$$A_x = \int_a^b xdy\;;A_{Y=}\int_a^b (x_1 - x_2)dy$$

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VOLUMES

$V_x = \pi \int_a^b y^2\,dx\,;\; V_{X=\pi} \int_a^b (y_1^2 - y_2^2)dx$

$V_y = \pi \int_a^b x^2\,dy\,;\; V_{y=\pi} \int_a^b (x_1^2 - x_2^2)dy$

SECOND MOMENT OF AREA

$I_x = \int_a^b r^2\,dA\,;\; I_{y=} \int_a^b r^2\,dA$

MOMENT OF INERTIA

$Mass = density \times volume$

$M = pv$

$DEFINITION: I = m\,r^2$

$GENERAL: I = \int_a^b r^2 dm = p \int_a^b r^2 dV$

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